

BEFORE THE FEDERAL COMMUNICATIONS COMMISSION

Washington, D.C. 20554

In the Matter of:)
Space Exploration Holdings, LLC) ICFS File No. SAT-LOA-20260108-00016
Application for Authorization of)
Non-Geostationary Orbit Satellite System)
(Orbital Data Center))

**SUPPLEMENTAL INFORMAL OBJECTION,
COMMENT AND CONSOLIDATED PRESERVATION OF
RIGHTS
OF NICKOLAI G. BAKKEN**

Filed Pursuant to 47 C.F.R. Sections 25.154(b) and 1.41 as a Supplemental Informal Objection and Public Comment; with Request for Waiver to the Extent Necessary Pursuant to 47 C.F.R. Section 1.3; with Preservation of Environmental Petition Rights Pursuant to 47 C.F.R. Section 1.1307(c); with Assertion of Party-in-Interest Status and Hearing Designation Rights Pursuant to 47 U.S.C. Sections 309(d) and 309(e); with Invocation of First Amendment Petition Rights and Pro Se Pleading Standard; and Incorporating by Reference All Prior Filings in This Docket in Their Entirety

Date: March 15, 2026

PRIOR FILINGS INCORPORATED BY REFERENCE

Informal Objection of Nikolai G. Bakken Filed: March 6, 2026 ICFS File No. SAT-LOA-20260108-00016 (Bakken IO)

Petition for Environmental Review and Supplemental Informal Objection of Nikolai G. Bakken Filed: March 9, 2026 ICFS File No. SAT-LOA-20260108-00016 (Bakken EP)

All supplemental preservation instruments and companion documents filed during the pleading cycle in this docket are incorporated herein by reference in their entirety and remain independently pending before the Commission.

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SUMMARY OF THE FILING

Petitioner identifies twenty-two facial discrepancies in the certified record of this application that independently require its return, and that collectively raise substantial questions about the candor of the certified submission and the basic fitness of the Applicant for the authorization it seeks.

Section I opens not with a legal argument but with a story, offered in the third person and presented as illustrative rather than sworn, because a proceeding the Applicant itself describes in civilizational terms deserves to be understood in the full human context in which it unfolded. That context includes extraordinary access barriers, a credentialing system undisclosed in the public notice, a filing infrastructure resolving to a commercial vendor domain rather than a verified .gov portal in apparent tension with the DOTGOV Online Trust in Government Act of 2020, a form instruction page that returns no instructions, a pleading type selector that adjusts nothing, and a silent filing system that provides no confirmation of receipt or conformity. A man navigated all of it, from Risaralda, Colombia, without counsel, in seven fewer days than the Commission allowed, while the Applicant received acceptance within five days of a filing the Commission's own system flagged as deficient. He makes music with his daughter from fast radio burst recordings. Their favorite film is WALL-E. He raises both not as sentiment but as evidence that the observational commons this application would degrade is a living resource with present human value, and that his daughter is still forming her understanding of what belongs to humanity and what does not.

Section II places that story in the federal context in which it occurred. Four distinct accountability frameworks were operative throughout the pleading cycle: the governance of orbital artificial intelligence infrastructure, for which no congressional authorization has been identified; a presidential executive action and existing statutory frameworks requiring disclosure and reporting to the All-domain Anomaly Resolution Office, whose intersection with a million-satellite orbital architecture the Commission must address or formally decline to address; a declassification mandate running concurrently with the pleading cycle, cited as a marker of the broader accountability climate in which this disposition will be rendered; and ongoing FOIA litigation, *Burleigh v. FCC*, No. 1:25-cv-01268 (D.D.C.), in which the Commission produced no text messages covering a period when the Applicant's controlling principal held a Special Government Employee designation and DOGE-affiliated personnel with disclosed financial interests in the Applicant's trust-controlled companies were embedded within the Office of the Chairman, with what was produced characterized by the presiding court as vague and uninformative. These four do not exhaust the matters of grave public importance this record surfaces but are not developed further because this Commission is not the forum equipped to fully resolve them. Each is preserved for the forum that will.

Section III develops the scientific paradox the Applicant's own vocabulary creates. Kardashev published his 1964 framework to describe listening for cosmological signals, not building orbital infrastructure. Fast radio bursts, arriving now for the first time in human history into instruments barely old enough to be called mature science, are electromagnetic signals of

exactly the character Kardashev described. FRB 121102, localized to a host galaxy three billion light-years distant, is barely nine years old as a subject of organized scientific inquiry. In the United States, a person under eighteen is a minor entitled to protection because the full scope of their potential has not yet had time to manifest. By that measure, this signal is a scientific child, arriving precisely within the framework Kardashev built for recognizing signals of cosmological origin. The section develops the sail analogy, the hardening paradox, and the genuinely novel engineering possibility of signal-transparent rather than signal-blocking node architecture, and closes with the observation that a civilization actually ascending the Kardashev scale cultivates its electromagnetic environment rather than saturates it. The array that could be a dual-purpose listening and energy-harvesting instrument of civilizational scientific value is, as currently filed, the system the science fears.

Section IV brings the peer-reviewed scientific literature that entered the public record during the pleading window to bear on the certified environmental assertion. Research published in a peer-reviewed geoscience journal in early 2026 documented stratospheric metallic accumulation from satellite reentries at rates that grow nonlinearly with constellation size. A 2023 body of work demonstrated measurable stratospheric contamination at constellation sizes orders of magnitude below one million satellites. A third study quantified mass flux upon atmospheric demise in ways that expose the gap between individual-satellite debris models and aggregate atmospheric reality. A fourth identified thresholds beyond which cumulative atmospheric effects become difficult to reverse. None of this literature is consistent with a certified assertion of no significant environmental impact on a record containing no bounded satellite mass, no material composition data, and no atmospheric loading projections. One day after the public notice for this application was released, a federal aerospace safety review documented what happens when the distance between a certified technical representation and operational reality is not closed before deployment. The Commission is not required to repeat that lesson. It is required to have learned it.

Section V presents twenty-two structural observations about the certified record, condensed and numbered i through xxii, each closing with the same fork: adjudicate on the record as filed or acknowledge that any cure is a material amendment requiring new public notice. Grouped across five categories: on the technical record, the certified data describes three satellites while the narrative requests one million, the system has no authorized downlink pathway, a mandatory validation error was accepted within five days during a partial government shutdown, the processing round waiver removes comparative evaluation protections, international coordination data is absent, and the primary function may fall outside the Commission's statutory jurisdiction. On ownership and fitness, the disclosures predate a merger that changed the corporate structure and left the majority of equity unattributed, the signatory's expert background removes any inference of inadvertence, no delegation letter establishing signatory authority appears on the record, and character questions follow the Applicant regardless of this filing's fate. On environmental integrity, the certification rests on absent data directly contradicted by peer-reviewed literature, and peer-reviewed research documents

unintended electromagnetic radiation from the existing constellation in ITU-protected radio astronomy bands at levels thirty-two times higher than the first generation. On notice and process, the environmental petition pathway was suppressed, the Applicant filed the first comment on its own application and appears to have uploaded the notice instrument itself, the acceptance date field is unpopulated, and the ICFS system's post-certification modification vulnerability is identified as a systemic question affecting every certified filing. On systemic risk, the waiver package dismantles the anti-speculation framework, and Petitioner formally opposes each pending waiver request individually and collectively. Across all categories, the deficiencies are filing-stage predicates, not post-grant conditions. There is no path to grant that does not first pass through the public.

Section VI opens with Hedy Lamarr, whose frequency-hopping spread spectrum patent expired before recognition and whose technique became an antecedent of the wireless commons this application proposes to saturate at civilizational scale. The section proceeds to the moral observation that the most important testimony is given from positions of pressure and cost, and calls the Commission to account for a stress test of institutional independence in which the standard the Applicant invoked and the record it produced are not consistent with one another. It closes with the acknowledgment that SpaceX built a genuinely transformative service the Petitioner knows from direct personal experience, and the observation that exercising authority to require a complete record is not opposition to the vision but the condition on which the vision can be built to last.

Section VII returns to what Kardashev actually wrote and develops the convergence that this proceeding has placed on the record for the first time: Lamarr patented a method for embedding structured information in apparent noise; Kardashev described a framework for detecting structured information arriving from cosmological distances; communications by wire and radio are now arriving from billions of light-years away; and artificial intelligence architectures capable of processing what they carry exist for the first time in history. These four facts have never coexisted before. A system that harvests solar energy in orbit while simultaneously capturing cosmological signals and feeding them into orbital compute solves three problems at once. The filed system goes in the opposite direction, saturating the medium it should be harvesting. The section incorporates the Elon Musk Milken Institute remarks of May 6, 2024, and the FRB 121102 scientific record, and closes with the observation that when a bounded, correctly certified communications system proposing dual electromagnetic harvest comes before this Commission, it should receive fast-track processing and public interest waivers. This is not that application. But the record now preserves what such an application would offer.

Section VIII formally preserves all threshold defects for the administrative record, petition for reconsideration, and judicial review. It invokes the SpaceX reusable rocket standard: the same applicant that built reusable rockets did not achieve that by asking the FAA to accept unbounded performance projections. Build it right, or do not build it yet. It closes with the jurisdictional observation that raw orbital compute generation is not communications by wire or

radio, that the harvesting and transmission of cosmological signals is communications in the most precise sense the Kardashev framework imagined, and that this distinction is not an invitation to append a communications purpose as a concessionary add-on. A signal that traveled three billion years to reach our instruments is not asking permission to be heard. The Commission must answer whether it will protect the conditions under which that signal can be received, or authorize the degradation of the medium through which it must pass.

Section VIII.A sets forth eight categories of relief requested: designation for evidentiary hearing under 47 U.S.C. Section 309(e) with a concurrent order to show cause, citing DA 26-76 as binding parallel precedent adopted eight days before the signatory executed the certification at issue here; abeyance under the Jefferson Radio doctrine as applied in DA 26-76; return as unacceptable for filing under 47 C.F.R. Section 25.112, simultaneously and independently of the hearing designation; a declaration that any submission of material not on the current record constitutes a material amendment under 47 C.F.R. Section 25.116; issuance of a new comprehensive public notice independently generated through the Commission's standard acceptance workflow, with parallel .gov-hosted access, an ECFS submission pathway, written confirmation to all parties, and acceptance of all prior submissions without prejudice; direction to the Office of Managing Director to establish a parallel docket entry on official .gov-hosted infrastructure; express findings on each of the twenty-two observations, each request in the concurrently filed Petition to Deny, and the pending environmental petition; and denial of any withdrawal not accompanied by dismissal with prejudice, with an express statement that character and fitness questions survive and attach to the Applicant in all concurrent and subsequent proceedings.

Section IX preserves all rights, standing theories, and participation rights to the fullest extent permitted by law, identifies X Corp and the February 4 and March 11 platform posts as material facts requiring express Commission determination on their ex parte status, incorporates all prior filings and all comments in this docket by reference, and closes with the request for leave to cure any identified deficiency before any adverse action is taken.

Section X sets forth the full nature, scope, and procedural posture of this filing, invokes the First Amendment right to petition, the liberal pro se pleading standard, the Noerr-Pennington doctrine, and the Commission's general waiver authority, and makes clear that three rights are not in dispute regardless of any procedural classification the Commission assigns: the right to petition, the right to comment in a noticed proceeding, and the right to request waiver for good cause. They are invoked. They are discernible. They are not waivable by implication, by label, or by silence.

Section X.A explains why this filing is deliberately unsworn on two independent grounds: the Commission's rules do not require it, and the state of the Applicant's certified record makes it logically impossible to frame a sworn objection with the specificity perjury liability demands against a record that is internally irreconcilable on its face. Every observation in Section V is a facial defect verifiable from the Commission's own records without reference to any testimony from this Petitioner. The evidentiary foundation is established in the sworn

instruments filed concurrently and previously. The indeterminacy that makes a sworn objection impossible to frame is not a deficiency in the objection. It is a deficiency in the Application.

The filing closes with a SHA-256 cryptographic hash published at secondary permanent public locations, because the Applicant's own certified record was modified after certification, ICFS is hosted on a non-government domain, the first Public Notice entry appears uploaded by the Applicant itself, and document integrity on this docket cannot be presumed.

PREFATORY NOTE

This document is a supplemental public comment and consolidated preservation instrument, not a sworn declaration, not a complete statement of all grounds for relief, and not a waiver of any argument, theory, or right preserved in any prior filing or in any filing incorporated herein by reference. The nature, scope, and procedural posture of this filing are set forth in full in Section X. All prior filings of this Petitioner in this docket, all supplemental preservation instruments filed during the pleading cycle, all comments and petitions filed by any party in this proceeding under ICFS File No. SAT-LOA-20260108-00016, all filings in the Electronic Comment Filing System that reference this docket or the SpaceX Orbital Data Center application by name or file number, and all filings in any ICFS or ECFS proceeding that cross-reference the ODC application or the issues raised herein, are incorporated by reference into this filing in their entirety to the extent necessary to preserve any argument, theory, ground for relief, or participation right that any such filing raises or fairly implies. Nothing in the classification of any such filing as a comment, informal objection, petition, or other vehicle limits the scope of this incorporation. The public record of this proceeding is treated as a unified evidentiary and argumentative whole for purposes of this filing and all subsequent proceedings in which this filing is cited or relied upon.

I: CITIZEN'S RECORD: A PARABLE OF SORTS

Before the administrative record begins, consider a story. Not because the story is necessary to the legal arguments that follow, because it is not, those arguments stand on the docketed record alone, but because a proceeding of this scale, one that an applicant has described in the vocabulary of civilizational transformation, deserves to be understood in the full human context in which it unfolded. The story that follows is offered in the third person. It is illustrative. Nothing in it is sworn testimony, and nothing in it should be read as such.

Nearly everything about the way this proceeding was structured made participation difficult for anyone without institutional support, and the man at the center of this story had none. He is an American citizen, one of approximately 330 million, traveling in Risaralda, Colombia, with a laptop and an internet connection provided by the same corporate family now asking this Commission to authorize a system one million times larger than what its own certified data describes. He pays for that service as needed to connect in remote signal areas, both domestically and internationally. It has, in his documented experience, been the infrastructure through which he communicated with the outside world during periods of personal danger and sought emergency assistance when no other option was available. His reliance on it is not theoretical. His interest in this proceeding is not abstract. He makes music using artificial intelligence tools, and together with his daughter they have used recordings of fast radio bursts as source material, feeding three-billion-year-old signals from a galaxy on the other side of the universe into compositional software and listening to what emerges. They have watched auroras, tracked eclipses, and seen the streaks of orbital debris burn through the atmosphere on nights that cannot be repeated. Their favorite film is WALL-E, which they have watched together more times than either can count, and which his daughter understood before she could fully articulate why: it is a story about a small, resilient machine left behind to steward what everyone else abandoned, singing music from the 1960s with a cockroach while a more sophisticated system searched the galaxy for signs that life could return home. The film does not condemn technology. It insists on stewardship before escape. A civilization that masters interplanetary travel while neglecting planetary care has not progressed. It has perfected the machinery of its own abandonment. The man in this story does not raise this as sentiment. He raises it because the Commission is being asked to authorize a system that would permanently degrade the observational commons through which signals like the ones he and his daughter turned into music must travel, and because his daughter is still forming her understanding of what belongs to humanity and what does not.

It happens that this man shares his given name, phonetically if not orthographically, with Nikolai Kardashev, the Soviet astrophysicist who in 1964 constructed the framework the Applicant chose to invoke in the narrative portion of this application. Kardashev built his civilization-scale rubric not as a commercial roadmap but as a scientific instrument for listening. He wanted to know whether electromagnetic signals from technologically advanced civilizations could be detected across interstellar distances, and whether the energy characteristics of those signals might reveal

how far along the development arc a civilization had traveled. His entire framework rested on an assumption so fundamental he did not state it explicitly because it seemed too obvious to require stating: that the electromagnetic environment would remain sufficiently clear of artificial noise for a listening civilization to hear the signal when it came. The man in this story noticed the shared name. He noticed what it implied about the proceeding he was about to enter.

Confronting that proceeding required, first, finding it. He went to the standard electronic system that the agency's public-facing website directs the public to use. The application was not listed there. He tried the backup email address published in official guidance. The server did not recognize the destination. He contacted the accessibility point listed in the notice itself and several days later was directed back to the system he had already confirmed was not functioning for this purpose. This consumed seven days, roughly a quarter of the thirty-day window the Commission made available for public participation in a proceeding that, if granted, would authorize the largest single expansion of human-operated orbital objects in history. Seven days. A quarter of the window. Lost to the Commission's own infrastructure. What began as an attempt to leave a constructive comment became something else entirely. Each barrier he cleared revealed another behind it. Each question he resolved surfaced two more the notice had not disclosed. The deeper he looked, the less the record resembled what the notice had described, and the more the proceeding revealed itself as extraordinary.

Keyed at last into a credentialing system whose requirement had not been disclosed in the notice, having navigated it through independent research alone, he accessed the docket at the end of those seven days. He read the certified technical instrument. He read the narrative. He found that the certified data described three satellites anchored to an orbital reference date nearly two years before filing, while the narrative described a system of up to one million satellites spanning altitude ranges between five hundred and two thousand kilometers. He found a mandatory validation error visible in the agency's own review system. He found ownership disclosures that predated a corporate merger introducing new investors and foreign-affiliated capital into the entity seeking authorization. He found an environmental certification of no significant impact on a record containing no bounded satellite mass, no material composition data, and no atmospheric loading projections. He found no evidence that the international coordination data required before any spectrum authorization may issue had been submitted or identified in the record. He found that upload records visible in the Commission's own filing system are consistent with the attachment having been uploaded by or on behalf of the Applicant rather than by Commission staff independently.

Only then did filing begin. But before a single word of substance could be placed on this record, the filer first had to understand what filing even meant in this context. The public notice, DA 26-113, released February 4, 2026, directed interested parties to file comments and petitions in the Commission's International Communications Filing System, in accordance with section 25.154 of the Commission's rules. That was the entirety of the procedural guidance provided to

the public. It did not explain that section 25.154 governs informal objections and that the procedural consequences differ materially from those of a formal petition to deny. It did not explain that an informal objection places different obligations on the Commission, different evidentiary standards on the filer, and different standing requirements on both. It did not explain that environmental objections must be raised through a separate procedural mechanism entirely distinct from the general comment pathway, and that a general comment expressing environmental concern, however well-supported, does not trigger the Commission's obligation to produce a substantive environmental review. It did not explain how waivers work, what the governing standard requires, how to formally oppose a waiver request, or how to preserve that opposition for judicial review. It did not fully identify the four separate waiver requests the Applicant had filed, each carrying its own standard, its own consequences if granted, and its own implications for parties who failed to oppose it on the record.

Locating the docket required a discovery the notice did not prompt and the system's own architecture obscures. An interested party who navigates naturally to www.fcc.gov finds a submission form to a general ECFS docket that does not correlate to the ICFS proceeding. Frustrated citizens were witnessed submitting comments to unrelated dockets. The ICFS CORES system, upon login, redirects users to a separate portal, and the docket becomes accessible only through both portals to a menu item labeled "Recents," a chronological feed with no prompt directing the public to use it, and search on both ECFS and ICFS sometimes supporting fuzzy search by name, other times not, without disclosing express use of a trailing "*" the public must discern to invoke it.

A member of the public directed to submit a document carrying criminal certification liability does not encounter a single departure from official federal infrastructure. The departure is layered. The authentication portal resolves to fcc-ext.okta.com/app/servicenow_ud, a domain operated by Okta, Inc. (NASDAQ: OKTA), a publicly traded commercial identity management company. Having authenticated through that commercial domain, the filer is routed into the filing system itself, which resolves to fccprod.servicenowservices.com, a domain operated by ServiceNow, Inc. (NYSE: NOW), a separate publicly traded commercial enterprise software company. Neither domain carries the .gov designation that the Commission's own website tells the public to look for when determining whether they are interacting with an official government system. The filer is not submitting credentials to the United States government. The filer is entering personal identifying information into one private corporation's authentication infrastructure and depositing certifiable filings into another private corporation's document management platform. The credential chain and the document chain are held by two different commercial entities, on two different commercial domains, with no government-domain intermediary visible at any point. This places the authentication record and the filing record for every party in this proceeding in the custody of third-party vendors whose obligations run to their contract with the Commission, not to the parties filing through the system or to the public relying on the integrity of the docket. The DOTGOV Online Trust in Government Act of 2020

established the .gov domain program so the public could identify official government digital services. OMB Memorandum M-23-10 (February 8, 2023) requires executive branch agencies to use .gov or .mil domains for official communications, information, and services. ICFS is not a social media account. It is not a collaboration tool the agency uses alongside the general public. It is a permit system. Public records indicate the ServiceNow-hosted ICFS launched in mid-2025, well after M-23-10 took effect. This infrastructure condition operates as a force multiplier for every other access barrier described in this section: the absence of instructions is more consequential inside a system the filer had reason to question before entering it, and the absence of confirmation is more consequential when the document was uploaded to a commercial domain whose relationship to the official record is nowhere explained. For a proceeding in which the docket presents structural conditions all parties can independently verify, including an unpopulated Public Notices acceptance date field, the Applicant's upload of the first pleading entry, and a public notice appearing as an Applicant attachment, the absence of .gov infrastructure means no party relying on this record can point to the government's own domain-level assurance that the chain of custody from submission to docketing occurred within government-controlled infrastructure. That hesitation is not paranoia. It is the response the DOTGOV Act was enacted to make unnecessary, and it belongs on this record as an infrastructure condition the Commission created and did not disclose.

Inside the system, having located the proceeding, a filer seeking to understand what their submission must contain navigates to the Pleadings and Comments section. At the top of that interface, adjacent to a Print Form option, sits a link labeled See Instructions. Selecting that link produces, in its entirety: "No Form Instructions exist for Subsystem(PAC) and Filing Type(PAC)." The link exists. The prompt exists. The instructions do not.

Despite the absence of instructions, the pleading type selector presents Petition as an available designation. Selecting it triggers no dynamic adjustment to the form, surfaces no standing requirements, no specificity standards, no affidavit or certification criteria, and no prompt directing the filer to the external rules governing whether the submission will be treated as what it has been designated. A first-time filer proceeding without counsel selects Petition, uploads a document, and presses Submit, with no qualification screen and no system-level acknowledgment that the selection carries requirements the submission may or may not meet. Such filings exist in this docket. Beyond the filing type itself, undisclosed requirements govern what a valid submission must contain: electronic signature standards, font and formatting constraints, verification requirements, service obligations, waiver procedures, and the distinction between curable and jurisdictional deficiencies. The notice disclosed none of them. The system prompts for none of them.

Additionally, the Pleadings and Comments interface itself carries a disclosure that the notice did not: a statement that filing a pleading through ICFS does not satisfy the service requirements, and that if Commission rules require service on other parties, the filer must serve them separately

by mail or as otherwise specified. Researching mail vs. email requirements was itself another maze. These are material procedural obligations whose existence was not mentioned in DA 26-113. A filer who read the notice, followed its instructions, uploaded their filing to ICFS, and believed they had completed the filing process would have no basis from the notice alone for knowing that numerous separate rule obligations affect acceptability. The disclosure that does exist appears only after the filer has already navigated the credentialing system, located the proceeding through the recency feed or arrived at the filing interface. It does not appear in the document that initiated the proceeding and directed the public to participate.

No document in this proceeding is publicly readable without credentials. Attachments uploaded to the ICFS docket are not viewable without CORES credentialing. This applies without distinction to every document in the record: the Applicant's own certified technical instrument, the Applicant's narrative, the Applicant's ownership disclosures, the Applicant's waiver requests, and every comment, objection, petition, and supplemental filing submitted by any party during the pleading window. A member of the public directed by DA 26-113 to file in ICFS and wishing first to read the application they were being invited to comment on would need to have completed the credentialing process that the notice did not mention, through the non-government domain infrastructure the system is hosted on, before the foundational document of the proceeding would be readable to them in any form. The record the Commission assembled through its public notice is not a public record in any ordinary sense of that phrase. It is a credentialed record, accessible only to those who discovered, navigated, and completed a registration process whose existence DA 26-113 did not disclose. The man navigating this system is not, by any reasonable measure, technologically unsophisticated. He builds complex digital workflows professionally and navigates novel technical environments as a matter of routine. The barriers he encountered were not the kind that yield to persistence or skill. They were structural, present from the first moment of participation, and they compounded with each procedural layer he uncovered.

In the end, having researched the rules independently through inference rather than disclosure, identified the correct vehicles by working backward from statutory text and Commission precedent the notice did not cite, drafted the arguments, assembled the required elements, determined through independent research what a valid signature requires in an electronic filing before an administrative body, and uploaded the documents through a credentialing system on a domain that does not carry the markers of official federal infrastructure, he received a digital receipt and acceptance on the public window with no confirmation that the filing was fully readable. No acknowledgment that it conformed to the general provisions of the applicable rules. No indication of whether it would be treated as carrying the procedural weight of the vehicle he had selected or would be silently reclassified into a category carrying less. The filing entered a silent state. It sits in that state still, dependent on its conformity with rules that were never disclosed to the person filing it, potentially subject to being disregarded by either party without direct reply, without acknowledgment, and without any obligation on the Commission's part to

explain why. The totality of that disenfranchisement left the man being asked to swear to ambiguity at every layer, a condition that only compounded once he reviewed what the docket actually contained. By the time he had fully read the certified record and understood what the agency's own infrastructure had and had not done, he had become so disenfranchised by what he found from both the Applicant and the Commission that he could no longer with any discernible logic participate faithfully under oath. Later sections of this commentary documents the impossibility in full: there are no fixed factual predicates in the Applicant's certified record against which a sworn statement could be framed with the specificity that perjury liability demands.

Every procedural obstacle described above was one the citizen navigated alone while an applicant whose certified filing contained encoding errors visible on its face, errors the Commission's own review system flagged as mandatory validation failures, received acceptance within five days of submission. That acceptance was publicly acknowledged in DA 26-113, released the same day it was processed. The proceeding was publicly noticed. ICFS notified the applicant immediately upon submission whether the filing had been received and assigned a file number. The applicant knew within days that its deficient filing had been accepted, publicly noticed, and placed into active processing. The citizen knew nothing of the kind, and still does not, except what he can infer from the fact that the documents appear in the non dot-gov system and that no party has told him otherwise.

Less than ten formal opposition filings in this docket preserve any procedural rights in the sense that administrative law requires for hearing designation, judicial review, or mandatory Commission response. Two of those were filed by private citizens proceeding without counsel. The man in this story is one of them. More than fifteen hundred public comments arrived during the same window, submitted by members of the public who linked their filings to the docket by number without ever having seen the docket itself. The credentialing barrier that gates access to every document in the record means that most of those commenters never read the Applicant's certified technical instrument, never reviewed the waiver requests they were implicitly being invited to address, and never encountered the structural conditions on the docket that all credentialed parties can independently verify. It is reasonable to assume that at least a portion of them, had they seen what the record actually contains, would have filed differently. But the docket's pleading form does not inform them what the agency is scrutinizing. It does not identify what the public should review to give the Commission the decision-making input a proceeding of this magnitude requires. The vast majority did not know that the environmental petition pathway existed, because the notice did not disclose it. Those voices represent astronomers, researchers, dark-sky advocates, rural users, and communities whose relationship to the night sky is ceremonial, cultural, and navigational as well as scientific. They were heard into a procedural channel that treats their concerns as boilerplate citizen commentary subject to categorical exclusion, rather than the pathway the notice failed to disclose that would have given those same concerns institutional weight. They suffered the same informational deficit as the man in this

story, on a smaller scale and with less recourse. If this story were entirely true, it would be extraordinary. The Petitioner offers it as illustrative. The Commission is invited to draw from it whatever it finds useful for understanding why a proceeding of this character demands more than a form disposition. The regulatory record that follows is not a parable. It is docketed, documented, historical and preserved, and it raises questions the Commission must answer expressly.

II: HOW FOUR FEDERAL FRAMEWORKS CONVERGE

Simultaneously with the filing of this application, and throughout the entire pleading cycle during which the public was invited to participate, at least four distinct and active federal accountability frameworks were operative in ways that directly implicate the corporate family of the Applicant, the governance questions the Applicant's own narrative raises, or the orbital domain in which the proposed system would operate. This application does not exist in a quiet moment of federal administration. It was filed on January 30, 2026, accepted within five days during an overlapping government shutdown, and publicly noticed on February 4, 2026. The Commission is identified here not as the forum for resolving all four frameworks, but because its obligation to render a reasoned, reviewable disposition does not diminish simply because the questions surrounding a proceeding exceed its own jurisdictional scope. Where the Commission's authority overlaps with these frameworks, it must act within that authority. Where it does not, it must say so. Silence in the face of a preserved question is itself a reviewable agency action.

Orbital AI compute infrastructure is the first framework and the one falling most squarely within the Commission's jurisdiction. The Applicant describes the proposed system's primary function as providing computing capacity to power advanced artificial intelligence, with radio frequency communications serving only backup and housekeeping purposes. The Commission is being asked to authorize an orbital AI compute platform on a certified record that contains no AI governance framework, no alignment standards, no safety disclosures, no human oversight architecture, and no deterministic command-custody protocols, for a system described as large-scale orbital AI compute infrastructure. During the same pleading cycle, the federal government was publicly adjudicating what governance obligations attach to AI providers of far smaller scale, including designating at least one AI company a supply chain risk to national security. The question of whether the Commission has explicit congressional authorization to license a system described primarily as AI computing infrastructure rather than communications is not a question this Commission can answer by implication or grant. It presents a matter of first impression requiring express congressional delegation, and this filing preserves it as such. The Commission's valuable contribution to that larger conversation is the record it builds here.

Layered directly onto the orbital domain in which this application proposes to operate is the second framework: a Presidential Executive action issued during the active pleading cycle

directing the release of government records relating to unidentified anomalous phenomena, operating alongside existing statutory frameworks that explicitly nullify nondisclosure agreements preventing reporting to designated oversight bodies including the All-domain Anomaly Resolution Office. These are not fringe references. They are a presidential directive and enacted federal law, and they bear on the orbital domain in which this application proposes to operate. A one-million-satellite constellation operating continuously in low Earth orbit would affect the ability of defense and intelligence communities, including AARO, to track, characterize, and investigate anomalous phenomena in that domain. The Commission should address, or formally decline to address, whether it has any oversight responsibility to AARO or to the frameworks that office administers, and whether Commission records are responsive to the disclosure obligations that law creates. That question ultimately belongs to a forum equipped to evaluate the intersection of national security, space domain awareness, and regulatory licensing. This Commission's authority overlaps with it in the limited but important sense that the orbital architecture this application would authorize is directly relevant to the space domain awareness concerns those frameworks address. The Commission need not resolve the question. It must acknowledge it exists on this record and state whether it has any responsibility to AARO or does not.

Ongoing pandemic accountability produces the third framework, cited here not as a substantive FCC matter but as a marker of the broader institutional accountability climate in which this disposition will be rendered and reviewed. Legislation established a declassification mandate with a defined window for the Director of National Intelligence to produce intelligence assessments regarding the origins of the COVID-19 pandemic, a window that ran concurrently with the pleading cycle of this proceeding. The Commission is not the forum in which those questions are resolved. But the principle that they cannot be foreclosed by agency silence applies here as it does elsewhere, and the Commission's disposition of this application should acknowledge, not ignore, the accountability climate in which it is rendered.

Most directly implicating the Commission's own decisional independence is the fourth framework. In *Burleigh v. Federal Communications Commission*, No. 1:25-cv-01268 (D.D.C.), FOIA litigation established that the Commission produced no text messages in response to requests for communications between DOGE personnel and FCC leadership during the period when the Applicant's controlling principal held a Special Government Employee designation and DOGE-affiliated personnel with disclosed financial interests in Applicant's trust-controlled companies were embedded within the Office of the Chairman. What was produced was characterized by the presiding court as vague and uninformative. The completeness and independence of the Commission's communications during the preceding regulatory period are disputed, and those disputes implicate this docket. The Commission must state on this record whether it can certify that its evaluation of this application was conducted independently of any influence arising from those relationships. That is a binary question with no implied answer, and each possible answer carries documented consequences this record cannot avoid. The

Commission must either make the finding or acknowledge that it cannot, and each answer carries documented consequences this record cannot avoid. These four frameworks do not exhaust the matters of grave public importance this proceeding has surfaced. Additional concerns, including questions touching on export control compliance, federal contracting certification accuracy, conflict-of-interest screening during the advisory service period, the gap between public representations about this program's regulatory viability and the deficiencies documented on this record, and the orbital domain awareness implications for classified national security programs whose oversight this Commission does not possess the mandate to conduct, have been identified and placed on the record in prior filings incorporated here by reference. They are not developed further in this forum because this Commission is not the forum equipped to fully resolve them. Each is preserved to the fullest extent permitted by law and will travel with this proceeding to whatever forum is necessary to obtain its resolution.

III: RADIO SILENCE AND THE KARDASHEV PARADOX

On the question of framing, the Applicant made its own choice. It described the proposed system as a first step toward a Type II civilization in the framework of Nikolai Kardashev. That choice is not rhetorical decoration in an otherwise technical filing. It describes a system whose primary purpose is declared, in the Applicant's own certified narrative, to be something other than telecommunications. It also invites examination measured by the scientific tradition Kardashev actually founded, rather than the commercial tradition the Applicant wishes to occupy. The Commission should take the Applicant at its word, because the Applicant chose the word.

Neither commercial nor infrastructural in its original intent, Kardashev's civilization scale was built to answer a single question about listening: whether electromagnetic signals produced by distant civilizations could be detected by a civilization at our level of development. His framework assumed, necessarily, that the electromagnetic environment in which that listening occurred would be clean enough for a signal to be distinguished from noise. The entire project of radio astronomy rests on that assumption. At the scale the Applicant proposes, the assumption is not under pressure. It is inverted.

Fast radio bursts are among the most energetic and scientifically significant transient phenomena in observational astronomy. They are brief, extraordinarily intense pulses of radio energy arriving from cosmological distances, originating from sources ranging from several hundred million to several billion light-years from Earth. They arrive carrying information about the intergalactic medium they traversed, the magnetic fields they passed through, and possibly about the processes that generated them. The question of whether any component of the fast radio burst population carries structured information rather than purely astrophysical signatures is not settled. It is open. The possibility that sufficiently advanced analytical tools applied to sufficiently clean observational data could distinguish structure from noise is the direct

methodological descendant of the reasoning Kardashev used when he argued that a sufficiently advanced civilization would leave an observable electromagnetic signature.

Artificial intelligence has fundamentally altered the analytical capacity available to radio astronomers. Machine learning architectures can now process telescope data at throughputs and pattern-recognition depths no prior generation could access. Several research groups are already applying machine learning to existing burst datasets and finding candidate structure that earlier analysis missed. The interference environment created by a million-satellite constellation does not merely degrade current observations. It forecloses the analytical future science cannot yet fully describe, because it produces a permanently elevated noise floor across precisely the frequency ranges where these signals arrive. The Applicant provided individual satellite-level analysis for a system it chose to describe as three satellites. The aggregate analysis for a million-satellite system does not exist on this record.

The youth of this science is precisely what makes the foreclosure so irreversible. Fast radio bursts were first identified in 2007. The entire scientific literature is less than two decades old. The field has had fewer years than many of the satellites currently in orbit. Authorizing a constellation that would permanently degrade the radio frequency environment in which the first generation of purpose-built detection infrastructure operates is a decision about whether a science still in its first generation of organized inquiry will be permitted to reach maturity.

A further sharpening of the stakes came with the discovery of repeating fast radio bursts. The source designated FRB 121102 was first reported in 2014, confirmed as a repeating source, and in 2017 pinpointed to a host galaxy three billion light-years away, making it approximately twelve years old as a known scientific object and barely nine years old as a localized phenomenon. In the United States, a person under eighteen is a minor, entitled to protection because the full scope of their potential has not yet had time to manifest. By that measure, this signal is a scientific child, arriving precisely within the framework Kardashev built for recognizing signals of cosmological origin.

Locked onto their repeatability, these sources offer something a single burst cannot: foreseeability. A repeating source is schedulable. It allows coordinated multi-telescope observations, longitudinal study of how burst characteristics change between events, and structured datasets that machine learning architectures are best positioned to exploit. Supervised training on a dataset with known temporal structure and repeated signal characteristics is a more tractable analytical problem than searching an archive of one-off events for buried pattern.

Embedded in what fast radio bursts represent, at minimum, is data, and data is by its nature communications which the Commission's governing framework addresses. The human genome was once a dataset too large to read. Protein folding was a problem too complex to model. Each transformation happened not because the data changed but because the tools available to process it crossed a threshold. Fast radio burst archives are a dataset of exactly that character. The

repeating sources are the scheduled witnesses. The question is not whether their testimony contains something important. The question is whether the environment in which future tools will attempt to hear it will still be clean enough to listen.

Think about how sails work. A sail does not generate wind. It intercepts what is already moving through the environment and converts its kinetic energy into directed force. A sufficiently large and correctly oriented array in orbital space could operate on analogous logic with respect to both electromagnetic radiation and the data it carries. A dual-purpose array designed to harvest solar energy while simultaneously functioning as a distributed radio telescope would represent a genuinely pioneering convergence. This is not proven engineering. It is a frontier concept whose value would depend entirely on maintaining the electromagnetic cleanliness of the orbital environment. An array that saturates the very spectrum it could be harvesting has chosen commercial throughput over scientific pioneering at the first moment in history when the tools to exploit that option are actually available. Large language models are not merely pattern recognizers. At sufficient scale, they are hypothesis generators. Applied to a sufficiently clean archive of fast radio burst observations, they could distinguish structured recurrence from astrophysical noise and generate testable predictions no human researcher working alone could produce. A permanently elevated noise floor does not merely slow that collection. It ends it, for the altitude bands and frequency ranges the proposed constellation would occupy, for as long as the constellation operates. A civilization actually ascending the Kardashev scale does not degrade its electromagnetic environment. It cultivates it. Kardashev measured civilizations by their capacity to harness energy. Electromagnetic radiation is energy. The spectrum is not a byproduct of advancement. It is the resource. Every natural electromagnetic signal arriving in the orbital environment, solar radiation, fast radio bursts, cosmic microwave background, magnetospheric emissions, is harvestable energy and harvestable information simultaneously. No prior human energy transition has presented this duality: wood, coal, oil, nuclear, and terrestrial solar each involved extracting energy from a medium that was not also the medium through which information traveled. The electromagnetic spectrum is both. It carries the energy and it carries the data. A civilization that grasps this builds infrastructure that captures both. A civilization that saturates the spectrum with unintended emissions from a million shielded nodes while invoking the name of the scientist who said we should be listening is not ascending the scale. It is descending it. The evidence of an advancing civilization is not the scale of its infrastructure. It is the clarity of its electromagnetic environment. An advancing civilization is one whose orbital architecture makes the universe more readable, not less. The system the Applicant has filed does not do that. The system the Applicant's own vocabulary demands would. The Commission has the authority to require it, and more than that, the mandate. The Communications Act charges this agency with managing the electromagnetic spectrum in the public interest. Managing does not mean allocating it until it is consumed. It means stewarding it so that it remains productive. A spectrum that carries cosmological signals into instruments that can read them is a spectrum producing value. A spectrum drowned in unintended emissions from a million shielded nodes is a spectrum in decline.

Hardening presents the third dimension of this analysis that the current record entirely ignores. The conventional response to space weather vulnerability in orbital electronics is shielding components through conductive enclosures that function as Faraday cages. But a hardened satellite is, by definition, a satellite that blocks the electromagnetic environment surrounding it. A constellation of one million hardened nodes is a constellation of one million localized suppressors of the very signal environment that radio astronomy, space weather monitoring, and fast radio burst detection depend on reading clearly. The two engineering imperatives, protect the hardware and preserve the signal environment, are in direct tension at this scale, and that tension has not been analyzed, modeled, or even acknowledged in the certified record before this Commission. A constellation whose failure mode during a Carrington-class event has not been described cannot be evaluated for public interest purposes on a record this sparse.

Implicit in this paradox is a genuinely novel engineering possibility the current application has not explored. The array needs the science it would degrade. It needs clean radio frequency capacity to monitor its own electromagnetic environment. It needs undisturbed stratospheric chemistry data to model its own reentry consequences. It needs space domain awareness infrastructure whose effectiveness depends on a less cluttered orbital environment than the array would create. If the array's electromagnetic architecture were designed around controlled signal transparency rather than shielding, the relationship between the array and the signal environment it inhabits could be inverted. Whether nodes of this character could simultaneously perform compute functions, contribute to distributed radio telescope arrays, and reduce rather than amplify aggregate interference is an open engineering question. It points toward a design in which the array and the observational commons are complements rather than adversaries. That design has not been described in the current application. The possibility that it could exist is preserved here, and as a genuine invitation to the Applicant to consider whether the system it is trying to build could be re-conceived as the system the science requires, rather than remaining, as currently filed, the system the science fears.

Returning to the Kardashev framework with which the Applicant opened its narrative: Kardashev's scale was not a blueprint for building. It was a framework for hearing. The Commission is being asked to authorize a system described in that vocabulary in a way that would permanently degrade the electromagnetic environment through which the signal Kardashev was actually writing about must travel. That irony belongs on the record expressly. The Commission need not resolve it. It must acknowledge it.

IV: INTERFERENCE SCIENCE DURING PLEADING WINDOW

Among the most consequential silences in the certified record before this Commission is the absence of any aggregate environmental analysis. During the pleading window of this proceeding, peer-reviewed scientific literature bearing directly on the environmental certification the Applicant filed entered the public record. That literature cannot be squared with the certified

assertion of no significant environmental impact, and the Commission must account for it before acting.

Measuring the atmospheric consequences of satellite reentry at constellation scales approaching what the Applicant proposes, research published in a peer-reviewed geoscience journal in early 2026 documented that metallic compounds, including aluminum oxide, released during satellite demise accumulate in the stratosphere at rates that grow nonlinearly with constellation size. The radiative consequences of that accumulation, including stratospheric ozone chemistry interactions and albedo effects, were identified as poorly characterized at proposed scales and as requiring urgent further study before large-scale constellation expansion proceeds. A separate body of peer-reviewed work published in 2023 demonstrated through direct measurement that stratospheric aerosol particles collected at altitude already contain metallic signatures consistent with spacecraft reentry at constellation sizes far smaller than what the Applicant proposes. That work documented measurable stratospheric chemistry contamination from populations orders of magnitude below a million satellites.

A third body of research quantified the mass flux of space debris upon atmospheric demise, documenting the gap between debris assessment models built on individual satellite parameters, which is the only form of analysis in the Applicant's certified record, and the aggregate atmospheric reentry chemistry observed at altitude. A fourth peer-reviewed study examined the long-term trajectory of satellite reentry consequences under current growth trends and identified thresholds beyond which cumulative atmospheric effects become difficult to reverse. Taken together, these four bodies of work describe a scientific landscape that did not exist when the environmental certification was filed and that the Commission cannot now ignore.

Before this literature entered the record, the Commission's environmental certification for this application had already been publicly noticed. The environmental petition filed by this Petitioner on March 9, 2026 placed that science before the Commission during the pleading cycle. The Commission must now account for it. A certified assertion of no significant environmental impact, filed on a record containing none of the bounded data necessary to evaluate aggregate atmospheric loading, cannot be sustained against peer-reviewed literature documenting measurable contamination at existing, far smaller constellation sizes. The certification must either be defended on the merits against that literature, or the Commission must acknowledge that the literature now on the record constitutes a substantial change in the evidentiary basis for the environmental posture of this filing, one that requires a fresh opportunity for public scrutiny before any authorization may issue.

In the day following the public notice for this application, on Feb 5th, NASA published the findings of its independent review of the Boeing Starliner Crew Flight Test, documenting the sequence of events in which a crewed spacecraft's propulsion system performed inconsistently with its certified parameters in ways that were not identified until the system was already

deployed in orbit. It is cited here not because spacecraft propulsion and satellite licensing share a regulatory framework, but because it is a contemporaneous, documented, federal record of exactly what happens when the distance between a certified technical representation and the system it purports to describe is not closed before deployment. Any authorization the Commission grants for this system is entirely downstream of the launch vehicles that would deliver it to orbit, vehicles subject to their own certification regime administered by a separate agency whose practices are themselves under active federal review, evaluating a different certified instrument filed by a different division of the same corporate family, during the same pleading window. The application now before this Commission describes a certified technical instrument encoding three satellites and a narrative requesting authority for one million. That distance is not a rounding error. It is not a formatting limitation. It is a choice, and the federal record of what follows from such choices was published one day after this proceeding was publicly noticed. The Commission is not required to repeat that lesson. It is required to have learned it.

Foreclosure of the scientific commons, degradation of the atmospheric environment, and the gap between certified representations and operational reality are not three separate problems on this record. They are three expressions of the same underlying deficiency: a certified record that does not describe the system being proposed, evaluated against an environmental baseline that does not exist, on a notice that did not disclose the pathway through which environmental concerns could have been formally preserved by the fifteen hundred members of the public who tried to raise them. The Commission must address all three dimensions or leave each one preserved for the forum that will.

V: STRUCTURAL DEFECTS: TWENTY-TWO OBSERVATIONS

What follows are not abstract policy disagreements with satellite technology or with the Applicant's commercial ambitions. They are structural observations about the certified record as it currently exists, and the Commission must answer each the same way: adjudicate on the record as it stands, or acknowledge that any future submission curing what is missing constitutes a material amendment requiring a new period for public participation. There is no third option, and the Commission should not imply one.

i) Irreconcilable on its face is the foundational technical discrepancy from which every other deficiency flows. The certified structured data encodes three satellites anchored to an orbital reference date of April 1, 2024, nearly two years before filing. That date is the epoch from which every interference calculation, conjunction analysis, debris projection, and coordination assessment must be derived. The narrative requests up to one million satellites spanning five hundred to two thousand kilometers. These are descriptions of fundamentally different things, and the certified inputs describe a system three millionths the size of what the narrative requests. The application is either for three satellites, in which case the narrative is surplusage, or the

certified data does not describe the proposed system, in which case the certification is materially inaccurate and any conforming submission is a material amendment requiring new public notice.

ii) Lagging behind corporate reality, the certified ownership disclosures predate a merger that closed in February 2026, consolidating the corporate parent with a prominent artificial intelligence company and introducing new investors including entities with reported sovereign wealth and foreign-affiliated capital. The Section 310(a) and (b) analysis must trace indirect foreign ownership through the full corporate chain. The Ownership Attachment identifies one holder controlling 41.5% through a private trust whose beneficiaries are not identified, despite the mandatory requirement to disclose beneficiaries in cases of fiduciary control. The remaining 58.5% is held by persons whose identity, citizenship, and nationality are entirely unknown on this record. The Commission cannot verify the certified "No" to the 25% foreign-ownership question when the majority of equity is unattributed. That is a facial arithmetic impossibility.

iii) Lacking any authorized pathway to move data between orbit and ground, the proposed system is operationally orphaned. The Applicant concedes the network would route traffic through its existing constellation via optical inter-satellite links. The Gen2 authorization granted ISL authority for the Gen2 Starlink constellation's own operations, not as a general-purpose relay for traffic from a distinct network of one million compute nodes. No modification applications were filed. If authorized tomorrow, this system would have no lawful mechanism to downlink its data to Earth.

iv) Concealed within a generic satellite notice, the actual regulatory scope was not disclosed. The filing selects both research satellite and commercial FSS designations, but no basis for Space Research Service appears in the Narrative or Waiver Requests. The system depends on inter-satellite communications as its primary relay mechanism, yet Inter-Satellite Service was not selected. The Narrative describes Ka-band TT&C, yet Space Operation Service was not selected. The result is an authorization whose noticed scope covers only backup Ka-band FSS while the primary mission, petabit-scale optical compute networking, operates entirely outside what the notice described. Parties did not know the full scope because the notice did not tell them, nor did any .gov site verify it during the pleading window.

v) On a record containing no bounded satellite mass, no material composition, no propellant data, no projected annual reentry mass, and no aggregate atmospheric loading analysis, the Applicant certified no significant environmental impact. That is a conclusion insulated from scrutiny by the incompleteness of the record that was supposed to support it. Peer-reviewed literature placed on this record during the pleading window documents measurable stratospheric metallic accumulation from existing constellations at scales far below what the Applicant proposes, directly contradicting the premise that a system of this size can be so certified on a record this sparse.

vi) Mandatory validation errors do not ordinarily accompany applications accepted within five days. This application contained an unresolved mandatory validation error in the uplink beam section, visible in the agency's own review system. The form instructions state all errors must be corrected before submission. The Commission accepted and noticed it anyway, during a partial government shutdown. That accelerated acceptance is markedly inconsistent with the treatment of at least one contemporaneous application for a comparable concept, which provided a complete technical record and had not been accepted as of the close of the comment period.

vii) Processing round rules ensure that where multiple applicants seek the same limited resource, each is evaluated comparatively. A separate orbital data center application in the same frequency bands was filed during the same period and remains pending. Granting a processing round waiver here while the comparable filing continues through ordinary review removes comparative evaluation for both applicants.

viii) Lacking predicate coordination data that the international treaty framework requires before authorization, the record cannot support a grant on this dimension alone. That data is not on this record. At least one other party placed documentation, under confidential treatment, of coordination commitments already binding on the Applicant in overlapping bands whose consistency with the proposed operations the Commission should evaluate before acting.

ix) Exceeding established jurisdiction, the Applicant's narrative makes clear the primary function is providing computing capacity for advanced AI, with RF communications serving only backup and housekeeping. The Ownership Attachment describes a different business: "provider of satellite broadband services." The question is whether the Commission may license a system whose own narrative describes communications as secondary, with orbital AI computing as the primary function, without explicit congressional authorization. The Applicant's civilizational framing sharpens rather than softens that threshold question.

x) The indeterminate nature of the proposed system forecloses any public participant's ability to identify the full range of harms a grant would produce. By certifying three satellites while requesting one million and declining to bound any parameter, the Applicant has made it impossible for any member of the public to evaluate what they are objecting to. A participant cannot waive objection to an injury the Applicant's own choices have made impossible to identify. The scope of preserved rights extends to all reasonably foreseeable consequences of a system at the scale the narrative describes, because that is the only scale this record provides.

xi) Expertise in Commission practice removes any inference of inadvertence. The Form 312 was certified by an individual whose background includes service as Senior Legal Advisor to an FCC Commissioner, service in the office of an FCC Chairman, service as Policy Advisor to the Chief of the Wireless Telecommunications Bureau, and service as Chief Counsel for the Communications and Technology Subcommittee of the House Energy and Commerce Committee during a period when the processing requirements governing this instrument were

under active consideration by that subcommittee. That individual does not appear in the Ownership Attachment. No board resolution or delegation letter establishing signatory authority appears on the record. Certifying the application as true, complete, and correct while simultaneously requesting waiver on the ground that the filing vehicle cannot accurately describe the system are not positions that coexist without producing a consequence the Commission must address. Either the certification is accurate and the application describes three satellites, or it is not, which is a material factual question requiring a hearing. Either answer excludes quiet inaction.

xii-1) Speculation of this scale is precisely what the surety bond was designed to deter. The Applicant seeks waivers of the bond, structured data requirements, processing round rules, milestone requirements, and related provisions. Waiving the bond for the most unbounded application the Commission has ever received inverts the purpose it was adopted to serve. Waiving disclosure requirements for an applicant with years of operational experience deploying large constellations allows substitution of an assertion of future information for the information the framework requires now.

xii-2) Petitioner therefore formally opposes each pending waiver request individually and collectively and requests denial of all of them. No deficiency identified in this filing or any prior filing may be cured through waiver. A waiver excuses compliance with a rule. It does not excuse compliance with the information the rule was designed to produce. The Commission cannot waive its way to a complete record. It can only receive one.

xiii) Orbital debris assessment is entirely a promise of future compliance rather than a present demonstration of safety. The analysis applies individual satellite-level thresholds without any aggregate analysis of what one million satellites would do to the collision risk profile of the orbital environment. At the described scale, the system would multiply actively controlled satellites in the affected altitude bands by a factor exceeding sixty, and the aggregate consequence has not been modeled on this record.

xiv) Leadership of this Commission made public statements during the active pleading cycle on a digital platform owned and operated by the corporate family of the Applicant following the February 2026 merger. The question is not whether a Chairman may comment publicly on a pending proceeding; it is whether doing so on infrastructure owned by the Applicant's corporate family creates an obligation the Commission must address on this record rather than by implication.

xv) Ongoing compliance with existing operations is a predicate to any new authorization, and the record does not confirm it. Peer-reviewed research in Astronomy and Astrophysics has documented unintended electromagnetic radiation from Starlink satellites between 110 and 188 MHz, including within an ITU-protected radio astronomy band. Second-generation satellites

emit this radiation at levels thirty-two times higher than the first generation, at intensities ten million times brighter than the faintest sources LOFAR observes. A Commission evaluating whether to grant a fleet sixty-seven times larger than the constellation already producing measurable interference in protected bands must account for the aggregate electromagnetic consequences.

xvi) More than seven days of the thirty-day window were consumed by the Commission's own infrastructure failures. The standard electronic system did not reflect this proceeding. The backup email address generated server failures. The accessibility contact referred the filer back to the inapplicable system. This chain affected at least one other independent filer who resorted to physical mail. Penalizing the public for delays caused by the Commission's own infrastructure is inconsistent with fair participation guarantees.

xvii) On February 4, 2026, five days after certification, DA 26-113 appeared as Attachment 6 in the Form 312 record under the Applicant's own attachment list. The certification covers "all statements made in this application and in all attached exhibits." Material added after certification is either uncertified or retroactively certified, which is not how federal criminal certifications operate. The certified record and the record in the Commission's system are not the same record. The threshold question is how the Applicant retained the ability to modify its certified docket record after certification. If the system permits post-certification modification without triggering recertification or a system-level lock, that is a systemic vulnerability affecting every certified filing, not merely this one. The ICFS record does not appear to show that the Commission independently generated the notice through its standard workflow. The first Pleadings and Comments entry, PC0107284, was also filed by the Applicant on the same date, classified as a "COMMENT" on its own application. The Commission must either confirm that it independently generated the notice, explain why the Public Notices tab and acceptance date field remain unpopulated if it did, or acknowledge that institutional separation was not maintained, in which case a new public notice must issue before any action may be taken.

xviii) No populated entry appears in the acceptance date field in the Commission's database, and no linked public notice appears where that entry should be. A contemporaneous application by a different company reflects all expected entries for a completed acceptance workflow. The Commission cannot rely on an unrecorded acceptance to advance this application as though standard prerequisites have been satisfied.

xix) Suppressed by the notice itself was the environmental petition pathway. The notice directed parties to file general comments without disclosing that satellite systems of this type are presumptively excluded from environmental review and that a specific petition must challenge that presumption. Roughly fifteen hundred members of the public were directed into a channel giving their environmental concerns less institutional weight than the undisclosed mechanism would have provided.

xx) The structured data requirement exists for a reason this application has rendered visible by ignoring it. The approved instrument, OMB Control Number 3060-0678, produces a machine-readable record exposed through a REST API enabling programmatic access and independent evaluation. The Applicant acknowledges the Schedule S "cannot accurately describe" the system. That is the problem, not the basis for excusing compliance. The Form Instructions identify an alternative REST API path in JSON format. The Applicant has in prior proceedings supplemented Schedule S with technical attachments and GIMS files. Here, for a system of far greater scale, it provided GIMS files for three satellites, no supplemental attachment, and no JSON submission. The deficiency is not software limitations. It is disclosure choices. Granting a waiver renders the API incapable of returning meaningful parameters and raises whether the Commission may excuse compliance with an OMB-approved collection without satisfying the PRA's requirements for material modifications.

xxi) Every threshold question on this record is an attribute of the Applicant, not merely of this filing vehicle. Under the Commission's character and fitness policy, these questions follow the applicant into all concurrent and future proceedings regardless of the fate of the filing that generated them. A withdrawal without a hearing abandons the vehicle while the certification questions remain unanswered. This Petitioner formally opposes any withdrawal not accompanied by dismissal with prejudice and reserves the right to file a timely opposition within the period the rules allow.

xxii) Most foundational of all: these deficiencies are filing-stage requirements whose satisfaction is a predicate to any lawful grant, not a consequence of one. The Commission cannot determine whether grant serves the public interest without knowing what system is being authorized, who owns the entity, whether environmental effects have been assessed on a sufficient record, and whether the coordination framework has been satisfied. Deferring to post-grant conditions would invert the regulatory framework, grant on an incomplete record, and deprive the public of participation in review of the operative information. Nor may deficiencies be cured by quiet supplementation. Any submission of updated ownership data, bounded parameters, aggregate environmental analysis, coordination data, corrected structured data, or any other absent material constitutes a material amendment under 47 C.F.R. Section 25.116 requiring new certification, new public notice, and a new comment period. On the record as it stands, the filing-stage predicates have not been satisfied. The application must be returned, dismissed, or designated for hearing before any action constituting a grant. There is no path to grant that does not first pass through the public.

VI: THE ACCOUNTABILITY MUST MATCH THE AMBITION

Preceding the wireless communications infrastructure this application proposes to expand by one million satellites, there was a patent. It was granted in 1942 to Hedy Lamarr, Hollywood actress and inventor, together with composer George Antheil. The patent described a frequency-hopping

spread spectrum communication technique designed to prevent radio-controlled torpedoes from being jammed by enemy forces. The patent expired in 1959 before it could be commercially exploited. The technology was not operationally implemented until the 1960s, by which point Lamarr had received no compensation. The Electronic Frontier Foundation did not recognize her contribution until 1997, three years before her death, by which point the frequency-hopping technique she and Antheil had patented without recognition had become an early antecedent of the spread-spectrum methods that underlie Bluetooth, military secure communications, and the broader wireless ecosystem this application proposes to saturate at civilizational scale. The spectrum this application proposes to occupy was made navigable in part by a woman whose name does not appear in any filing in this docket, whose patent generated no royalties, and whose contribution the institutions that benefited from it declined to acknowledge for half a century. The Commission is being asked to authorize an expansion of that architecture at civilizational scale. It should do so with full awareness of whose shoulders it stands on, and what it means to build civilizational infrastructure without accountability to the foundational contributors whose uncompensated work made it possible. Long before Lamarr and long before Kardashev, the deepest traditions of human moral life understood something about the relationship between institutional power and the voices that power tends not to hear. The great wisdom traditions, each in its own vocabulary, have recognized that the most important testimony is often given not from positions of comfort and institutional favor but from positions of pressure, exposure, and cost. There is a figure in those traditions, present across multiple streams of human moral inheritance, who speaks most clearly precisely when speaking is most costly, who does not abandon the record when the institution has turned away, and who understands that remaining present in a proceeding that has structurally disadvantaged your participation is itself a form of witness. The Petitioner does not claim that mantle. He observes only that the right to petition the government for redress does not require institutional support, legal credentials, or parity of resources. It requires only persistence and a record. He has provided both. He has provided them from Colombia, through a credentialing system the notice did not disclose, into a docket where the opposing party employs some of the most accomplished satellite lawyers in the country, on behalf of fifteen hundred members of the public whose voices were directed into a procedural channel that gives them less institutional weight than they deserved, through no fault of their own. The voice that speaks clearly when it is most costly to do so is the voice the record remembers. This record will memorialize what matters were before it.

Enormous ambition deserves enormous accountability, and the Commission has the institutional capacity to provide it. Nothing in this proceeding is foreclosed to the Applicant. The objections on this record do not argue that the technology is impossible, that the science is wrong, or that the vision is unworthy. They argue that the record through which that vision has been brought to this agency does not support the authorization being requested, and that building something of civilizational significance on a defective regulatory foundation serves no one, including the Applicant, because authorizations built on incomplete records are inherently vulnerable to the kind of sustained, documented, multi-track challenge this record now contains. SpaceX built a

genuinely transformative service. The existing constellation has demonstrated that satellite broadband can reach communities that no other infrastructure serves, including communities in conflict zones, in regions of natural disaster, and in the developing world where terrestrial alternatives do not exist. The Petitioner knows this from direct personal experience. The question before the Commission is not whether the Applicant is capable of building something remarkable. That answer is already on the public record of what has been built. The question is whether the application currently pending provides the evidentiary foundation on which the Commission can lawfully authorize what comes next. On the record as it currently stands, it does not, and the Commission owes the public, the Applicant, and the integrity of its own process the honesty of saying so clearly, along with a precise statement of what a complete record would need to contain. A fully bounded, correctly certified, properly noticed, internationally coordinated application for a system that could genuinely serve the public interest would look very different from what is currently on this docket. The Commission has the authority and the institutional knowledge to tell the Applicant exactly what that application would need to contain. Exercising that authority is not opposition to the vision. It is the condition on which the vision can be built to last.

What must also be said, because the moment requires it, is that the institutions implicated in this proceeding are being pressure-tested. The Commission's leadership made public statements during the active comment window on infrastructure owned by the Applicant's corporate family. In the months preceding this permit, the Applicant's controlling principal simultaneously held a federal advisory role with embedded access to non-public regulatory data. A certified filing containing a mandatory validation error was accepted within five days during a government shutdown, on a record fewer than ten formal parties have scrutinized, while fifteen hundred members of the public were directed into a procedural channel the notice did not explain. No allegation of impropriety is made. None is necessary. The Applicant raised the standard by describing this system in civilizational terms. An authorized representative with extensive experience in Commission proceedings, congressional oversight, and the professional responsibility standards governing practice before this agency then presented the Commission with a certified record that produces the material questions this filing documents. The standard invoked and the record produced are not consistent with one another, and the Commission must address that inconsistency before acting. Institutions that fail stress tests of this character do not fail all at once. They fail through a series of individually defensible small concessions that, read together in retrospect, describe an agency that confused proximity to ambition for participation in it.

Accountability is what the choices made in this proceeding will ultimately be measured against, because those choices will be read by the world the proceeding helps create. That is not a rhetorical flourish. It is an administrative fact. The Commission's disposition of this application will establish precedent for how the largest proposed expansion of human activity in the orbital domain is evaluated, who gets to participate in that evaluation, what information is required

before a grant may issue, and whether the public interest standard means something when applied to a civilizational-scale proposal or whether it yields to institutional momentum. The Petitioner has participated in this proceeding as exhaustively as a single unrepresented citizen can navigating barriers of the Commission's own creation, because the questions this record raises are too important to be left to fewer than ten formal opposition filings at the end of the comment window. The Commission now has the record those filings have built. The accountability this proceeding requires matches the ambition the Applicant has declared, or it does not. The Commission should ensure that it does.

VII: ON WHAT KARDASHEV ACTUALLY WROTE

Long before this application was filed, long before the word Kardashev appeared in a federal licensing proceeding, Nikolai Kardashev published his foundational 1964 paper under a title that most people who invoke his scale have never read: "Transmission of Information by Extraterrestrial Civilizations." Not infrastructure. Not computing capacity. Transmission. Information. Extraterrestrial. The paper was not a roadmap for commercial development but a scientific argument about detectability and contact. His central claim was precise: that a civilization sufficiently advanced to be classified at the higher levels of his scale would produce electromagnetic signatures of such distinctive character and such extraordinary power that a listening civilization at our level of development could, in principle, detect them across interstellar and intergalactic distances. He was not writing about what we might one day build. He was writing about what we might one day hear and harvest. The Applicant borrowed his vocabulary to describe a commercial filing. The Commission is now being asked to authorize that filing in a way that would permanently degrade the electromagnetic environment through which the signal Kardashev was actually writing about must travel.

Listening is the capacity Kardashev's entire framework assumes and requires. It requires clean instruments. It requires protected frequency ranges. It requires an electromagnetic environment in which a signal of cosmological origin could be distinguished from the noise produced by the civilization doing the listening. The entire scientific tradition that flows from his work, from the earliest systematic searches for extraterrestrial intelligence through the current generation of fast radio burst detection campaigns, rests on that foundational requirement.

Elon Musk, a principal of the Applicant, is on public record observing that he has never encountered an extraterrestrial and that no Starlink satellite has ever been required to maneuver around one. (Remarks at the 2024 Milken Institute Global Conference, May 6, 2024, reported Space.com, May 9, 2024; substantially repeated on the Peter Diamandis Moonshots podcast, January 6, 2026, clip affirmed by Musk on X the same date.) The remark is offered not as criticism but as an invitation to examine its premises. The fast radio bursts now arriving at Earth's observatories are, by the most precise scientific characterization available, exactly what Kardashev said we should be listening for. The signal designated FRB 121102 has been traveling

toward this planet for three billion years. It departed its source before complex animal life existed on Earth, crossed intergalactic space through the entirety of human evolution, and arrived at our instruments within the past decade, during the first window in which we had built instruments capable of detecting it. Whether that signal carries structured information from a technologically advanced source or is purely astrophysical in origin is a question science has not yet answered. What science has answered is that these signals are real, arriving from extragalactic distances, repeating on increasingly characterizable schedules, and carrying information no other observational tool can provide. No Starlink satellite has been required to maneuver around one yet. That is precisely because we have not yet built the listening infrastructure that would tell us whether we should. Here is what is being missed. Lamarr patented a method for embedding structured information in apparent noise. Kardashev described a framework for detecting structured information arriving from cosmological distances. Communications by wire and radio are now arriving from billions of light-years away. Artificial intelligence architectures capable of processing what they carry exist for the first time in history. These four facts have never coexisted before. The Applicant describes a system that harvests solar energy in orbit to power computation. That is half the harvest. The electromagnetic environment those satellites would inhabit carries cosmological signals encoding data that has traveled for billions of years. An array that harvests the energy and captures the data solves three problems at once: power, a dataset no terrestrial instrument can replicate, and direct feed into the orbital compute the Applicant already proposes to build. Solar collection plus signal capture plus onboard AI processing equals a system that listens, harvests, and learns. That is what ascending the Kardashev scale actually looks like. The filed system goes in the opposite direction, saturating the medium it should be harvesting, burning the fishery to fuel the boat. Every one of these threads converges on the same electromagnetic environment the Applicant proposes to degrade. That is the central scientific fact of this proceeding. The Commission's authority over the electromagnetic layer is not merely the authority to permit its use. It is the authority to require that its use enhance what the layer carries rather than degrade it, to condition orbital architecture on harvest rather than saturation, and to recognize that interstellar signals arriving into the bands this Commission administers are themselves a public resource whose reception the public interest standard protects. When a bounded, correctly certified communications system proposing dual electromagnetic harvest comes before this Commission, the Commission should embrace it, strongly entertain fast-track processing, and grant waivers that align with the public interest to accelerate its deployment. This is not that application. But the record now preserves what such an application would offer. Fifteen hundred voices wrote into this docket during the pleading window. The astronomers, the dark-sky advocates, the researchers, the rural users, the citizens who filed without knowing which procedural channel would carry their words. Read together, they are saying the same thing in different vocabularies: do not degrade the layer. Protect and enhance it. The signals are arriving. They have been traveling for three billion years. It is time to maneuver our arrays to greet them, not to drown them out for good with orbital compute shells.

VIII: SILENCE IS NOT RESOLUTION

Laid across all twenty-two observations presented here and in all prior filings incorporated by reference, the threshold defects identified in this proceeding are now formally preserved for the administrative record, for petition for reconsideration, and for judicial review. These observations are not the product of opposition to progress. They are the product of a conviction that progress of this scale demands rigor equal to its ambition, and that a civilization serious about ascending the scale it has invoked must be equally serious about the scientific commons that ascent requires. The objections on this record are protective of scientific children, of phenomena barely a decade old as localized study-worthy objects whose full potential cannot be foreclosed by infrastructure decisions made before the tools to realize that potential even exist. The same applicant that built reusable rockets did not achieve that by asking the FAA to accept unbounded performance projections and fill in the details later. It achieved it through diligent engineering. This proceeding asks for nothing more than the same standard. Build it right, or do not build it yet. Granting or continuing to process an application built on mathematically irreconcilable certified data, obsolete ownership records, an operationally orphaned system, missing coordination predicates, an unsupported environmental certification, and a notice instrument with documented procedural irregularities constitutes a departure from the reasoned administrative action the law requires of this agency in every proceeding, regardless of the identity or ambition of the applicant.

One further jurisdictional observation this record requires. The Commission's enabling authority extends to communications by wire or radio. The Applicant describes the primary function of the proposed system as orbital computing capacity for advanced artificial intelligence, with radio frequency communications serving only backup and housekeeping purposes. Raw compute generation, standing alone, is not communications by wire or radio in any ordinary reading of that phrase. The harvesting and transmission of cosmological signals, by contrast, is communications in the most precise sense the Kardashev framework imagined: the reception, across cosmological distances, of electromagnetic signals carrying structured information, and the transmission of that received data to ground-based analytical infrastructure. That function fits squarely within the Commission's jurisdiction. It is also the function the current application does not describe, does not propose, and does not design for. This observation does not invite the Applicant to append a communications purpose as a concessionary add-on. The Commission's Part 25 framework governs satellite systems whose communications purpose is primary, not incidental, not remedial, and not grafted onto an existing filing after the jurisdictional question has been raised. If communications is to be the basis for this authorization, it must be the foundation of the system as designed and certified, not an orphaned clause added to preserve access to a licensing framework the primary described function does not otherwise reach. A signal that traveled three billion years to reach our instruments is not asking permission to be heard. It is arriving whether this Commission acts or not. What is being asked is whether this agency will protect the conditions under which that signal can be received, or whether it will

authorize the degradation of the very medium through which the signal must pass. Fifteen hundred members of the public wrote into this docket. Seven independent parties filed formal oppositions. The Petitioner has been present in this forum, under institutional pressure, without counsel, for as long as the record required, because signals that have traveled that far and voices that have fought that hard to be heard on this docket share something in common: they both demand answer. The Applicant chose to invoke Kardashev. Having done so, it cannot refuse to engage with what Kardashev's framework actually requires, and neither can this Commission. Silence in the face of a signal that crossed the observable universe is not neutrality. It is a choice. Silence in the face of twenty-two documented deficiencies placed on this record by seven independent parties is not disposition. It is abdication. That is why this Petitioner is in this forum. That is why silence is not resolution. And that is why the Commission must answer expressly.

Now preserved to the fullest extent permitted by law, for petition for reconsideration, judicial review, review under the Administrative Procedure Act, and any other accountability mechanism permitted by law, are all categories, authorities, cases, treaties, instruments, and doctrinal bases identified in this document and in all prior filings incorporated by reference. The Commission's disposition of this Application, whatever form it takes, must address these preserved questions expressly or leave a reviewable record of its failure to do so. If the Commission declines any relief requested in this submission or in any prior filing, Petitioner requests that it state the basis in sufficient specificity to permit meaningful subsequent review. Nothing in this record, in the label attached to any prior filing, or in any procedural determination the Commission may make constitutes a waiver of any argument, theory, injury, right, or ground for relief fairly supported by this record. Silence is not resolution. Implication is not a finding. What is not addressed on this record is preserved for the forum that will address it. The night sky belongs to everyone. The signals Kardashev described have been traveling for three billion years waiting for instruments clean enough to receive them. Whatever the Commission decides, it should decide as though the decision will be read by the world the decision creates.

VIII.A: RELIEF REQUESTED

For the reasons set forth in this filing and in all prior filings incorporated by reference, Petitioner respectfully requests:

1. **Designate this proceeding for evidentiary hearing under 47 U.S.C. Section 309(e) and concurrently issue an order to show cause under 47 U.S.C. Section 312(c) why the Application should not be dismissed with prejudice.**

Section 309(e) is mandatory. Where substantial and material questions of fact are presented and the Commission cannot find that grant would serve the public interest, it "shall formally designate the application for hearing." On January 23, 2026, the Commission's Media Bureau issued DA 26-76, designating a hearing under these same provisions upon finding substantial and material questions of fact regarding unauthorized foreign control, misrepresentation and lack of candor in violation of 47 C.F.R. Section 1.17(a)(1), and character qualifications under the 1986 Character Policy Statement. The Bureau found "a potential pattern of indifference to, and exploitation of, the Commission's Rules that cannot be addressed with a forfeiture order, consent decree, or other lesser enforcement measure," made the Enforcement Bureau a party, and placed the burden of proof on Commission staff. The hearing issues in DA 26-76 parallel this record: ownership questions the certified record cannot answer, a certification structurally incompatible with the underlying facts, and threshold character-qualification concerns. The inference of inadvertence available to the parties in DA 26-76 is not available here. The authorized representative who executed this certification served as Senior Legal Advisor to an FCC Commissioner, in the office of an FCC Chairman, as Policy Advisor to the Chief of the Wireless Telecommunications Bureau, as an Honors Attorney in the Spectrum Competition and Policy Division, and as Chief Counsel to the congressional subcommittee with primary legislative oversight of the Commission and its statutes. That background does not permit the Commission to treat what is on this record as a clerical oversight. If those facts warranted hearing designation in a three-station broadcast proceeding, the same standard applies with at least equal force to the largest satellite application in Commission history. DA 26-76 was adopted eight days before that representative executed the certification at issue here.

2. **Hold the Application in abeyance pending resolution of the hearing, consistent with the Jefferson Radio doctrine as applied in DA 26-76.**

Under Jefferson Radio, "a licensee may not have anything to assign or transfer unless and until the pending character qualifications questions have been resolved." DA 26-76 applied this doctrine to suspend processing while the hearing proceeded. The same principle applies here. No technical cure, no supplemental submission, and no post-filing amendment should be accepted until the threshold questions placed on this record by seven independent parties have been resolved through the administrative process the statute provides.

3. Return the Application as unacceptable for filing under 47 C.F.R. Section 25.112, simultaneously and independently of the hearing designation.

The twenty-two structural observations in Section V each independently demonstrate that the certified record does not contain the inputs the Commission's own rules require. The record cannot support a public interest finding on the technical, ownership, environmental, or coordination dimensions this filing identifies.

4. Declare that any submission of material not on the current record constitutes a material amendment under 47 C.F.R. Section 25.116, requiring new certification, new public notice, and a new period for public participation.

This includes updated ownership data, bounded technical parameters, aggregate environmental analysis, international coordination data, corrected structured data, and any other information the twenty-two observations identify as absent. There is no path to grant that does not first pass through the public.

5. Issue a new, comprehensive public notice independently generated through the Commission's standard acceptance workflow, and in connection with that notice:

- (a) Publish the new public notice and all non-confidential application materials on the Commission's .gov-hosted website, accessible without ICFS credentialing.
- (b) Establish a parallel comment submission pathway through the Electronic Comment Filing System on fcc.gov, referencing the ICFS file number.
- (c) Accept all filings previously submitted through ICFS and ECFS as timely and properly filed, without prejudice for any deficiency attributable to the non-.gov infrastructure, the undisclosed credentialing requirements, or the absence of form instructions.
- (d) Provide written confirmation to all parties who have filed in this proceeding that their submissions have been received, are readable, and will be treated as carrying the procedural weight of the vehicle designated at the time of filing, or if the Commission intends to reclassify any submission, state the basis expressly and afford an opportunity to cure.

The Commission's own ICFS record shows that the first entry on this docket, PC0107284, was filed by Space Exploration Holdings, LLC on February 4, 2026, classified as a "COMMENT," described as "Form Attachment for public comment to the FCC on SpaceX's orbital data centers application." DA 26-113 also appears as Attachment 6 in the Form 312 record, uploaded five days after certification. The Commission's notice under Section 309(b) is an independent agency act. If the Commission cannot confirm on the record that the existing notice was the product of its own independent review, a new notice must issue before any action may be taken.

6. Make express findings on each of the twenty-two observations in Section V, each express-findings request in the concurrently filed Petition to Deny, and the pending Petition for Environmental Review filed March 9, 2026.

The Commission cannot act on this Application without addressing the environmental petition expressly. Silence on a pending petition is not a grant. It is not a denial. It is a preserved ground for review. Express findings are necessary to render the Commission's disposition reviewable under the Administrative Procedure Act and to ensure that threshold issues placed on the record by seven independent parties are not left to silent or implied resolution.

7. Deny any withdrawal not accompanied by dismissal with prejudice, and state expressly that the character and fitness questions on this record survive and attach to the Applicant in all concurrent and subsequent proceedings.

DA 26-76 found that the Jefferson Radio policy exists to prevent a party from "sell[ing] out from under a potential disqualification." The same principle forecloses withdrawal here. The Applicant cannot withdraw its way out of the record this proceeding has produced. The questions follow the Applicant into any refiled, modified, or successor application for this system or any system of comparable scale proposed by any entity under common ownership or control. This Petitioner will oppose any withdrawal request within five business days of its filing.

To the extent this filing is treated as an informal objection under 47 C.F.R. Section 25.154(b) and Section 1.41, Petitioner requests that it receive the full substantive consideration the Commission's rules provide for that vehicle, and that no relief requested herein or in any prior filing be deemed waived by the procedural classification the Commission assigns. All relief requested in the concurrently filed Petition to Deny, the Bakken IO filed March 6, 2026, and the Bakken EP filed March 9, 2026, is incorporated by reference and independently preserved. Nothing in this prayer narrows, supersedes, or forfeits any relief requested in any prior or concurrent filing in this docket.

Respectfully Submitted,

/s/ Nikolai G. Bakken

Nikolai G. Bakken

IX: PRESERVATION OF RIGHTS, STANDING, AND PARTICIPATION

Nothing in this record, in the label attached to any prior filing by the Commission, or in any procedural determination the Commission may make regarding the classification of any submission in this docket constitutes a waiver of any argument, theory, injury, right, or ground for relief fairly supported by this record. The Commission's disposition of this application, whatever form it takes, cannot be lawfully insulated from review by implying resolution of issues that have been expressly raised and not expressly answered. If the Commission declines any relief requested in this submission or in any prior filing, the Petitioner respectfully requests that the Commission state the procedural and legal basis for that denial in sufficient specificity to permit meaningful subsequent review. Silence is not resolution. Implication is not a finding. What is not addressed on this record is preserved for the forum that will address it.

X Corp, operating as the social media platform under common ownership with the Applicant's controlling principal following the February 2026 merger, is identified here not as a separate respondent but as a material fact in the administrative record of this proceeding. Commission leadership made public statements during the active pleading window on that platform. Those statements are either reportable ex parte communications that must be placed on the official record, or they are not, and the Commission must state which. The question is not whether X Corp's ownership connection to this proceeding is improper in isolation. The question is whether the record is protected against the consequences of that connection, and whether the Commission can certify that its evaluation of this application was conducted independently of any influence arising from the common ownership of the platform on which its leadership chose to comment on a pending proceeding involving that platform's owner. Petitioner preserves all rights to administrative and judicial review of any disposition of this proceeding. This preservation is affirmative, open-ended, and coextensive with the scope of the Application itself, which by its own terms declines to bound its operational parameters and therefore implicates an unbounded range of potential regulatory, environmental, public-interest, and legal consequences that no participant can be expected to enumerate in advance of their maturation. To the extent that any of Petitioner's filings are found to require a waiver of any procedural requirement, Petitioner hereby requests such waiver on the ground that the circumstances documented across this record, including the Commission's own infrastructure failures, the undisclosed credentialing prerequisites, the misdirection of official staff, and the absence of form instructions for the filing vehicle the notice directed the public to use, constitute extraordinary circumstances justifying relief under the Commission's general waiver authority. The barriers Petitioner encountered were not of Petitioner's making. They were of the Commission's making. A party cannot be penalized for failing to satisfy requirements that the agency's own systems and notices made effectively unknowable.

Continuing in full force and independently pending before the Commission are all prior filings in this docket, including the Informal Objection filed March 6, 2026, the Petition for Environmental

Review and Supplemental Informal Objection filed March 9, 2026, and all supplemental preservation instruments filed during the pleading cycle. Each is incorporated here by reference in its entirety. None is merged into, superseded by, or diminished by this supplemental comment. Nothing in this filing narrows, waives, or forfeits any argument, theory, ground for relief, injury allegation, or participation right asserted in any prior filing or preserved in any prior preservation instrument. Petitioner asserts standing and party-in-interest status on all grounds fairly supported by this record, including direct-user injury arising from ownership of a Starlink terminal and use of the Applicant's existing service on an intermittent and as-needed basis, aesthetic and environmental injury arising from established practices of night-sky observation and celestial study including practices undertaken with Petitioner's minor daughter, informational injury arising from the Applicant's decision to file a certified record that omits the bounded technical, ownership, environmental, and coordination data the Commission's filing framework is designed to place before the public, and procedural injury arising from the documented access barriers that prevented meaningful docket access for seven days of the comment period. No enumeration of standing theories in this or any prior filing shall be construed to waive, narrow, or forfeit any other theory of standing, injury, or participation right supported by the present or future record. All standing theories are pleaded in the alternative and are preserved to the fullest extent permitted by law, including theories that may become more fully articulable as the record develops, as the Applicant provides information currently withheld, or as governing legal standards evolve.

Obligation to designate a proceeding for hearing when the statutory predicate is met is mandatory, not discretionary, and Petitioner contends the predicate is plainly met here on multiple independent grounds including the identity of the system proposed, the ownership and control structure of the entity seeking authorization, the environmental consequences of the proposed operations, the procedural adequacy of the notice, the candor and completeness of the certified filing, and the character and fitness of the Applicant as bearing on all of the foregoing. To the extent the Commission determines that any aspect of the environmental record triggers the extraordinary circumstances standard under its own rules, Petitioner preserves the argument that the scale of the proposed system, one million satellites producing continuous reentry events across altitude bands whose aggregate atmospheric loading has never been modeled, objectively constitutes such extraordinary circumstances, and that the Commission's obligation to conduct a substantive environmental review before any authorization may issue is therefore not discretionary on this record. The independently pending environmental petition filed March 9, 2026 formally invoked that standard and remains unresolved. The Commission cannot act on this application without addressing that petition expressly. Silence on a pending petition is not a grant, and it is not a denial. It is a preserved ground for review.

Reserving all bases for petition for reconsideration of any Commission action in this proceeding and for judicial review of any final Commission action in the appropriate federal court, Petitioner expressly preserves the right to hearing designation, the right to oppose any withdrawal not

accompanied by dismissal with prejudice, and the right to file a timely opposition to any withdrawal request within the period the Commission's rules allow. The questions on this record do not disappear if the Applicant abandons the current filing vehicle. The Commission is requested to state expressly that the character and fitness questions already on this record survive any withdrawal and attach to the Applicant in all concurrent and subsequent proceedings before this Commission regardless of the filing vehicle in which they arose. This preservation encompasses review of any disposition reached by implication, including any disposition that treats preserved issues as resolved without expressly addressing them. An agency decision that does not address a material issue expressly raised in the record is itself a reviewable agency action, and Petitioner intends to treat it as such. To the extent that any filing deadline, vehicle classification, or procedural designation affecting Petitioner's submissions is found to require formal waiver, Petitioner requests that waiver now, on the ground that the combination of access barriers, undisclosed procedural requirements, and infrastructure failures documented across this record satisfies any threshold the Commission applies to extraordinary circumstances warranting procedural relief, and that denying such relief while simultaneously processing a deficient applicant submission on an accelerated timeline would be inconsistent with evenhanded administration of the Commission's rules.

Preserved to the fullest extent permitted by law, for petition for reconsideration under the relevant provisions of the Communications Act, for judicial review in the appropriate federal court, for review under the Administrative Procedure Act, and for any other administrative, legislative, or public accountability mechanism permitted by law, are all categories, authorities, cases, treaties, instruments, and doctrinal bases identified in this document and in all prior filings in this docket incorporated by reference to the extent necessary. The Commission's disposition of this Application, whatever form it takes, must address these preserved questions expressly or leave a reviewable record of its failure to do so. Silence is not resolution. Implication is not a finding. The issues preserved here will travel with this proceeding to whatever forum is necessary to obtain their resolution. The night sky belongs to everyone. The orbital environment is a shared commons held in trust for all present and future users of the Earth. The electromagnetic spectrum is a public resource whose allocation must serve the public interest as that interest is understood across the full breadth of human knowledge, value, and moral tradition. Hedy Lamarr's patent expired before the institution recognized what it owed her. The signals Kardashev described have been traveling for three billion years waiting for instruments clean enough to receive them. This Petitioner has been present in this forum, under institutional pressure, without counsel, from a city in Colombia, for as long as the record required. Whatever the Commission decides, it should decide as though the decision will be read by the world the decision creates. To the extent the Commission identifies any deficiency in any filing submitted by this Petitioner in this docket, whether procedural, formal, or substantive, Petitioner requests that the Commission provide notice of that deficiency and afford a reasonable opportunity to cure before taking any adverse action based upon it, consistent with the principle that unrepresented filers who navigated procedural barriers not of their making and not disclosed in

the notice that invited their participation cannot be held to a standard of technical perfection that the Commission's own infrastructure made impossible to achieve. This request for leave to cure is made in the alternative and does not concede that any deficiency exists.

X: NATURE, SCOPE & PROCEDURAL POSTURE

Presented here as a public comment and supplemental informal objection filed in an established federal administrative forum pursuant to the Commission's rules governing participation in international satellite licensing proceedings, this document is not a sworn declaration, certified statement, affidavit, or testimony offered under any evidentiary standard in any proceeding. No portion of this document, including its opening narrative, its scientific observations, its jurisdictional arguments, its philosophical and cultural references, its preservation instruments, or its closing statements, is offered as sworn testimony, and none of it is properly characterized as such or subject to strike on that basis.

Every portion of the illustrative narrative that opens this document is offered in the tradition of the amicus brief's contextual parable, the congressional witness's framing account, and the long-standing American practice of citizen commentary before administrative bodies that begins with the observation that a proceeding of this scale must be understood not only through its regulatory dimensions but through the human perspective it implicates. That narrative is presented in the third person. The Petitioner neither disclaims the events it describes nor certifies them in this forum. It is offered as illustrative, explicitly not as sworn testimony, and is therefore not properly characterized as witness testimony subject to strike.

The scientific observations in this document are offered as the informed perspective of a citizen participant who reviewed the docketed record and the peer-reviewed literature placed on that record during the pleading window. They are not offered as expert testimony and do not require that characterization to be preserved for the administrative record and for subsequent review. The jurisdictional arguments, preservation instruments, procedural observations, and closing statements are grounded in the docketed record, in the Commission's own published rules and instructions, and in publicly available legal authority. They stand independently of the narrative section and do not depend on it for their legal force.

Invoked here expressly and without condition are the First Amendment right to petition the government for redress, the liberal pro se pleading standard applicable to unrepresented filers, the protections applicable to good-faith participation in an established federal administrative forum, and the Noerr-Pennington doctrine as applicable to all statements made in the course of that participation. Nothing in this document is offered to supplement, modify, or clarify any sworn statement made elsewhere in this docket. Nothing in this document constitutes a waiver of any argument, theory, ground for relief, or participation right preserved in any prior filing. All

prior filings remain independently pending and are incorporated here by reference to the extent necessary.

To preserve full procedural effect for any and all filings submitted in this docket, Petitioner requests, to the extent necessary, a waiver of any procedural requirement, deadline, vehicle classification, format specification, or filing prerequisite that may apply to this or any prior submission, on the following grounds. First, the public notice that initiated this proceeding, DA 26-113, did not disclose the existence of the International Communications Filing System credentialing prerequisite, did not identify the environmental petition pathway as a separate and distinct procedural vehicle, did not describe the service requirements applicable to formal pleadings, did not identify the filing type requirements applicable to petitions, and did not provide form instructions, because the Commission's own system confirms that no form instructions exist for the applicable subsystem and filing type. Second, the Commission's own infrastructure produced documented failures during the initial seven days of the comment window, including the non-availability of this proceeding in the standard electronic comment system, server delivery failures on the published backup submission address, and official staff misdirection to an inapplicable system, all of which are extraordinary circumstances not of Petitioner's making and fully attributable to the Commission's own filing architecture. Third, Petitioner proceeded without counsel, from Risaralda, Colombia, navigating a procedural landscape that was materially undisclosed in the notice that invited participation, under conditions that would have made compliance with unstated requirements effectively impossible for any unrepresented filer exercising reasonable diligence. Fourth, the Commission cannot simultaneously accept for processing a deficient applicant filing containing mandatory validation errors on an accelerated five-day timeline while enforcing technical procedural requirements against an unrepresented objector whose barriers were caused by the Commission's own systems, without raising the precise arbitrariness concern that the Administrative Procedure Act's reasoned decision-making requirement is designed to prevent.

In the event any of Petitioner's submissions are characterized as comments rather than petitions, Petitioner requests that they be treated as informal objections carrying the full procedural weight available under the Commission's rules for that vehicle, and that they be evaluated on their substantive merits without prejudice for any procedural deficiency attributable to the barriers documented in this record. To the extent any submission requires elevation to petition status to preserve hearing rights, party-in-interest status, reply rights, response obligations on the Commission, or any other procedural entitlement, Petitioner requests that elevation now, with good cause shown by the totality of extraordinary circumstances documented across this record. To the extent any procedural maze remains wholly undiscernible at this stage of the proceeding, as it does, because the Commission's own filing infrastructure provided no instructions, no dynamic guidance, no qualification confirmation, and no acknowledgment of receipt that would allow an unrepresented filer to know whether their submission achieved its intended procedural effect, Petitioner preserves the right to supplement, clarify, or recharacterize any prior

submission as additional procedural pathways become identifiable, and requests that the Commission not treat the indiscernibility of those pathways as a basis for denying the substantive relief requested across all filings in this docket.

On this record, nothing in this section, and nothing in the label the Commission attaches to any filing in this docket, shall be construed to narrow, waive, or forfeit any argument, theory, procedural right, substantive ground for relief, or participation right fairly supported by this record. The procedural classification of this filing is a question for the Commission to resolve expressly, on the record, with a stated basis. It is not a question that may be resolved by implication, by silence, or by the unilateral application of requirements that the notice inviting participation did not disclose. To the extent that any rule governing the availability of waiver itself would otherwise operate to bar relief, including any requirement that a waiver request be filed by a party of a particular procedural classification or within a particular window not disclosed in the public notice, Petitioner requests waiver of that rule as well, narrowly and only to the extent necessary, on the ground that good cause and extraordinary circumstances have been shown across this record, and that a citizen petitioner exercising constitutionally protected petition rights in a proceeding whose procedural architecture was not disclosed in the notice that invited participation cannot be bound by the meta-procedural requirements of a waiver framework that was itself undiscoverable through the channels the Commission made available. The waiver requested is not broad. It is precisely coextensive with what is necessary to ensure that the substantive questions preserved across all filings in this docket receive the merits review that the public interest standard requires, and that the voice of an unrepresented citizen who navigated every barrier this proceeding placed in his path is not extinguished by the very labyrinth he successfully crossed.

No procedural classification this Commission assigns to this filing or to any prior filing in this docket can displace three rights that are not in dispute and are not subject to any undisclosed prerequisite, any credentialing requirement, any form instruction that does not exist, or any filing vehicle whose requirements were withheld from the public notice that invited participation. The First Amendment right to petition the government for redress is self-executing and requires no procedural vehicle to attach. The right to submit a public comment in a noticed federal proceeding is conferred by the notice itself and cannot be forfeited by infrastructure failures the Commission created. And the right to request waiver of any Commission rule for good cause shown, including the rule governing waiver itself, is expressly provided by the Commission's own published rules and requires only a showing of good cause and extraordinary circumstances, both of which have been made across this record with specificity and documentation that no responding party has contested. Petitioner invokes all three rights now, expressly, without condition, and without dependence on any procedural determination the Commission has not yet made. Everything else preserved in this filing rests on those three foundations. They are discernible. They are invoked. They are not waivable by implication, by label, or by silence.

Specifically incorporated by reference into this record, irrespective of any official Commission determination regarding their procedural classification, are the following communications made on the X platform through verified government accounts during the active pleading cycle of this proceeding: the post made on February 4, 2026 by Commission Chairman Brendan Carr on the date of the public notice release, which publicly characterized the SpaceX Orbital Data Center application and welcomed public engagement with its merits on infrastructure owned by the Applicant's corporate family; and the post made on March 11, 2026 by Chairman Carr during the active pleading window of this proceeding, which publicly characterized a formal statutory petition filed by another party in terms expressing a view on the merits of that petition, also on infrastructure owned by the Applicant's corporate family. Both posts were made during the period this proceeding was governed by permit-but-disclose ex parte rules as announced in DA 26-113. Both are incorporated not because their procedural classification has been resolved, but precisely because it has not. The Commission must determine whether those posts constitute ex parte presentations that must be disclosed and placed on the official record, whether they constitute public statements by a decision-maker raising adjudicative prejudgment concerns under governing administrative law standards, whether replies and interactions generated on the Applicant's platform infrastructure in response to those posts are similarly subject to preservation and disclosure obligations, and whether the use of the Applicant's corporate family's platform to solicit public engagement on a pending proceeding involving that corporate family triggers any obligation beyond permit-but-disclose treatment. Petitioner further preserves, pending any official Commission determination, whether public statements made through verified government accounts on X constitute comments within the meaning of the Commission's rules, ex parte presentations subject to mandatory disclosure, or communications of another character whose treatment must be stated expressly on the record before any disposition may issue. Each of these questions is binary. Each answer carries documented consequences this record cannot avoid by implication or silence. All are preserved expressly, both posts and all associated interactions are incorporated to the fullest extent the law permits, and the Commission is requested to address each question expressly in its disposition of this proceeding.

X.A: WHY THIS FILING IS UNSWORN

This filing is deliberately unsworn, for two independent reasons: the Commission's rules do not require it, and the state of the Applicant's certified record makes it logically impossible to frame with the specificity perjury liability demands. On the first ground, this filing is submitted as a supplemental informal objection and public comment under 47 C.F.R. Sections 25.154(b) and 1.41, neither of which requires verification, oath, or sworn declaration. The twenty-two structural observations in Section V are facial defects apparent on the Commission's own records and independently verifiable by Commission staff without reference to any testimony from this Petitioner. The certified structured data encodes three satellites; the Commission can confirm this from its own ICFS system. The validation error is visible on the Commission's own Schedule S Review page. The Ownership Attachment's date is printed on its face. The peer-reviewed studies

are published in journals of record. The Applicant's upload of the first entry in the Pleadings and Comments record is timestamped in the Commission's own system. None of this requires an oath to be true. It requires the Commission to look at its own records. The concurrently filed Petition to Deny carries a verification under 28 U.S.C. Section 1746. The Petition for Environmental Review filed March 9, 2026 carries a sworn declaration under the same statute. The evidentiary foundation for this proceeding is established in those sworn instruments. This supplemental comment provides the structural analysis, the preservation architecture, and the broadest procedural vehicle the Commission's rules make available. An informal objection that identifies facial defects from the Commission's own records does not require the filer's oath to obligate the Commission to address them before acting. It requires the Commission to read its own file.

On the second ground, a sworn statement requires the declarant to attest to specific facts against a fixed factual predicate with the precision that perjury liability demands. This record does not provide that predicate. The certified structured data describes a system of three satellites. The narrative describes a system of one million. The Applicant certified both as "true, complete, and correct" while simultaneously stating the filing vehicle "cannot accurately describe" the proposed system. The Ownership Attachment describes a corporate structure that ceased to exist two days before the public notice was released. The Commission's own ICFS system contains no evidence that the standard acceptance workflow was followed. The first entry in the Pleadings and Comments record appears to be the Public Notice uploaded by the Applicant itself. The Commission's own form instructions page confirms that no instructions exist for the filing type the notice directed the public to use. Against which version of this record would a sworn objection be framed? The three-satellite system the Commission's technical review tools can evaluate, or the million-satellite system no technical data on this record describes? The pre-merger entity the Ownership Attachment reflects, or the post-merger entity that actually exists? A citizen cannot swear to the specific harms of a system the Applicant has declined to define, against an ownership structure the Applicant has declined to disclose, in a proceeding whose own notice instrument the Applicant appears to have generated. The indeterminacy that makes a sworn objection logically impossible to frame with the specificity perjury liability requires is not a deficiency in the objection. It is a deficiency in the Application, and the responsibility for it belongs to the party that created it. To the extent the Commission or a reviewing court determines that the inability of a participating citizen to frame a sworn objection against an internally irreconcilable certified record constitutes an independent procedural defect in this proceeding, that determination is preserved for petition for reconsideration under 47 U.S.C. Section 405 and for judicial review under 47 U.S.C. Section 402(b).

Respectfully Submitted,

/s/ Nikolai G. Bakken

Nikolai G. Bakken

Pro Se

U.S. Citizen

Risaralda, Colombia

nikolai.bakken@gmail.com

Mailing Address:

200 104th Ave, PMB 321,

Treasure Island, Florida 33706

- END

SHA-256 DOCUMENT INTEGRITY

A SHA-256 cryptographic hash of this document as filed is published in a separately filed Certificate of Document Integrity on the original docket and at

<https://www.linkedin.com/in/nickolaibakken>

The hash was computed on the final version immediately before upload. Any party may verify integrity by downloading the document and computing the hash. This mechanism is filed because the Applicant's certified record was modified after certification and the first Pleadings and Comments entry was uploaded by the Applicant itself. The integrity of documents on this docket cannot be presumed, but this one can be verified.

This document consists of 44 pages.

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